

In linear transformations $T(\mathbf{x}) = A\mathbf{x}$, applying A to the standard basis vectors $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ yields the columns of A :

$$T(\mathbf{e}_1) = A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \text{Column 1 of } A, \quad T(\mathbf{e}_2) = A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \text{Column 2 of } A$$

1. $A_1 = \begin{bmatrix} 2 & 0 \\ 0 & 0.5 \end{bmatrix}$ (Scaling)

- **(a) Basis Vector Transformations:**

$$T(\mathbf{e}_1) = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad T(\mathbf{e}_2) = \begin{bmatrix} 0 \\ 0.5 \end{bmatrix}$$

- **(b) Visual Effect:** Stretches the image horizontally by $2 \times$ and compresses it vertically to $0.5 \times$.
- **(c) Rank:** $\text{rank}(A_1) = 2$ (Full rank, $\det(A_1) = 1 \neq 0$).
- **(d) Information Loss:** No information is lost because A_1 is invertible.
- **(e) Connection to Columns:** Column 1 scales horizontal distances by a factor of 2 along x , while Column 2 scales vertical distances by 0.5 along y .

2. $A_2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ (90° Counter-Clockwise Rotation)

- **(a) Basis Vector Transformations:**

$$T(\mathbf{e}_1) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad T(\mathbf{e}_2) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

- **(b) Visual Effect:** Rotates the image 90° counter-clockwise around the image center.
- **(c) Rank:** $\text{rank}(A_2) = 2$ (Full rank, $\det(A_2) = 1 \neq 0$).
- **(d) Information Loss:** No information is lost because A_2 is an orthogonal rotation matrix ($A_2^{-1} = A_2^T$).
- **(e) Connection to Columns:** Column 1 maps the horizontal unit vector onto the positive y -axis, while Column 2 maps the vertical unit vector onto the negative x -axis, executing a 90° counter-clockwise rotation.

3. $A_3 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ (Horizontal Shear)

- **(a) Basis Vector Transformations:**

$$T(\mathbf{e}_1) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad T(\mathbf{e}_2) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

- **(b) Visual Effect:** Slants the image horizontally to the right proportional to pixel height.
- **(c) Rank:** $\text{rank}(A_3) = 2$ (Full rank, $\det(A_3) = 1 \neq 0$).
- **(d) Information Loss: No information is lost** because A_3 is invertible ($\det(A_3) = 1$).
- **(e) Connection to Columns:** Column 1 leaves the x -axis undisturbed, while Column 2 shifts points horizontally rightward proportionally to their vertical distance y .

4. $A_4 = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$ (Reflection in y -axis)

- **(a) Basis Vector Transformations:**

$$T(\mathbf{e}_1) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \quad T(\mathbf{e}_2) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

- **(b) Visual Effect:** Flips the image horizontally across the vertical axis (y -axis).
- **(c) Rank:** $\text{rank}(A_4) = 2$ (Full rank, $\det(A_4) = -1 \neq 0$).
- **(d) Information Loss: No information is lost** because A_4 is self-invertible ($A_4^{-1} = A_4$).
- **(e) Connection to Columns:** Column 1 negates x -coordinates to reflect horizontal positions, while Column 2 preserves all y -coordinates unchanged.

5. $A_5 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ (Projection onto x -axis)

- **(a) Basis Vector Transformations:**

$$T(\mathbf{e}_1) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad T(\mathbf{e}_2) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

- **(b) Visual Effect & True Matrix:** Collapses the entire 2D image into a 1D line along the x -axis. For visualization in Python, a small scale parameter $\epsilon = 0.02$ can be used in place of 0 in the second row, but the **true matrix** remains $A_5 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.
- **(c) Rank:** $\text{rank}(A_5) = 1$ (Rank-deficient, $\det(A_5) = 0$).
- **(d) Information Loss: Dimension and information ARE lost.** The 2D image collapses into a 1-dimensional subspace, completely wiping out all vertical (y -axis) information irreversibly.

- **(e) Connection to Columns:** Column 1 preserves horizontal positions along the x -axis, while Column 2 zeroes out all vertical components, flattening the entire 2D image plane into a single line.